

Controlled five-seed experimental design

Paired seeds 40-44 · configuration-level comparisons

Locked reference comparison

Six RT-DETR configurations; YOLOv10 Base versus FAM
Full-data, fixed-budget training · final checkpoints · MtErie evaluation

VALIDATION-LED COMPARISONS

Stage A · five seeds for every configuration

Train: 3,123 paired frames · validation: 896 paired frames
Whole-video split · matched candidate/control comparisons

RT-DETR FAM: architecture, RCRA, box-guided and mixed consistency
RT-DETRv2 and YOLO26: Base versus FAM
Budget: 10 epochs for RT-DETR / v2; 50 epochs for YOLO26

Primary: best validation mAP@50 · minimum improvement 0.001
Secondary: final-epoch checkpoint · no early stopping

Aggregate promotion criterion
Mean paired $\Delta \geq +0.01$ and $\geq 4/5$ positive deltas
Plus configuration-specific checks

Not met:
no Stage B

Only if all criteria are met

Stage B · matched full-data confirmation

Fresh pretrained initialization · all 4,019 training pairs · five seeds
Fixed budget · final checkpoint only · no validation-based selection
Evaluation on the common 708 MtErie pairs after training is complete

Separate diagnostic and generalization evidence

Modality, alignment and replay audits · synthetic stress · other acquisitions